KL UNIVERSITY · ENGINEERING CAPSTONE PROJECT SPECIFICATION

DSCI-20 — Civic Open-Data Dashboard for Public Accountability

Canonical ID
DSCI-20

Dominant specialization
Data Science & Big Data Analytics

Technical domain
Data Science and Big Data Analytics

Strongest impact
Industry Impact

Reference source: data.gov.in · Open Government Data (data.gov.in)
https://data.gov.in/

READ THIS BEFORE CHOOSING THE PROJECT

Two-semester capstone project charter

CP1 + CP2 · complete engineering system

PROBLEM TO SOLVE

Governments publish open data — budgets, services, crime, health, education — but citizens and officials rarely have it in a usable, insight-driven form. The consequence is incomplete, incorrect or non-reproducible evidence reaching downstream decisions in Media, Entertainment & Social Platforms.

WHO BENEFITS AND WHAT OWNERSHIP IS ESTABLISHED

Primary beneficiaries are Open Government Data (data.gov.in) users, maintainers and technical reviewers; the cited source establishes context or source custody but does not identify an operational problem owner. No sponsor, funder or supporting organisation for the KL student implementation is stated. The KL deployment stakeholder and operational decision owner must be confirmed before approval.

FINAL ACHIEVEMENT REQUIRED

Deliver a civic open-data dashboard turning messy public datasets into honest, transparent, decision-useful insight.

REFERENCE TO REPRODUCE IN CP1

Analyse trends and regional disparities. This is the O2 reference against which every Civic Open-Data Dashboard for Public Accountability contribution must be compared.

DIFFERENTIATING CONTRIBUTION IN CP2

Analyse relationships (e.g., allocation vs outcomes) honestly. This is the O3 technical contribution that must be implemented, isolated and evaluated separately.

COMPLETE FINAL DEMONSTRATION

Build the public accountability dashboard with transparent methodology. The final demonstration must show the complete Civic Open-Data Dashboard for Public Accountability workflow rather than an isolated component.

INDEPENDENT ACCEPTANCE

Evaluate analytical honesty and dashboard usefulness. The result must be independently checked under representative, boundary and failure conditions.

WHY THIS IS CAPSTONE-WORTHY

This is a two-semester culminating project because the team must establish the O2 reference “Trends and regional disparities”, design and evaluate the O3 contribution “Relationships e.g., allocation vs outcomes honestly”, integrate the O4 system “Public accountability dashboard with transparent methodology”, and defend the O5 result under independent, boundary and failure conditions.

Problem evidence and approval condition

OFFICIAL CONTEXTUAL EVIDENCE REQUIRING LOCAL PROBLEM VALIDATION

Evidence locator: Government open datasets · **Open cited source**

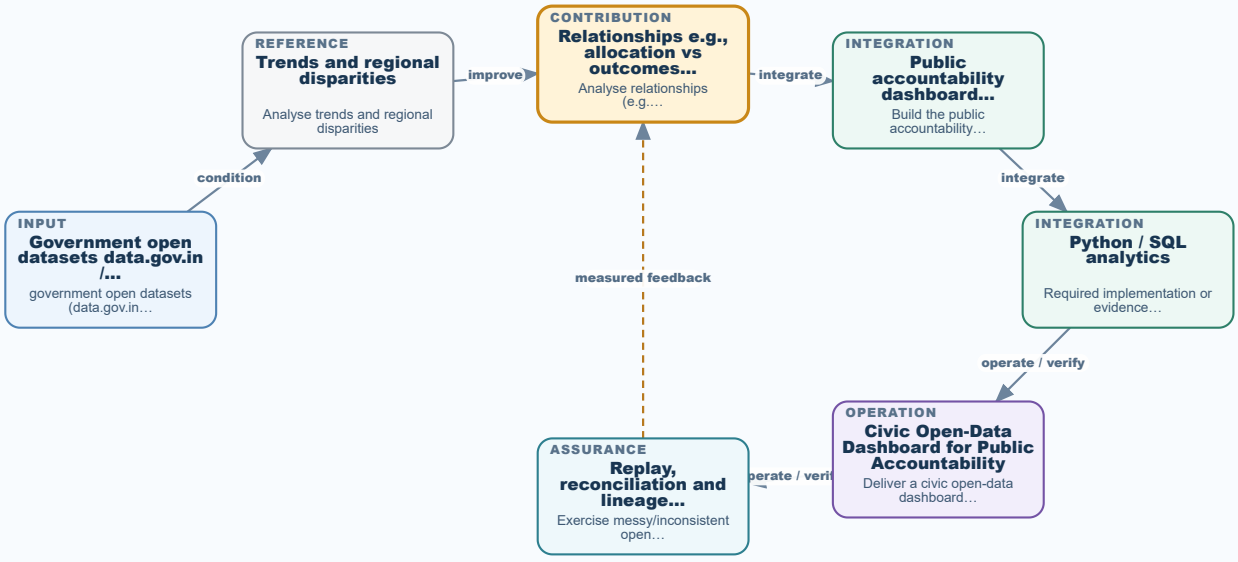
What the source establishes: The cited source establishes the engineering context, technology or application area; it does not by itself prove that the exact local problem remains unresolved.

What the team must verify: Before implementation proceeds, reproduce the stated limitation with versioned evidence or attach a signed problem-validation record from a named stakeholder. If neither is achieved, replace or rescope the project.

Pictorial project specification

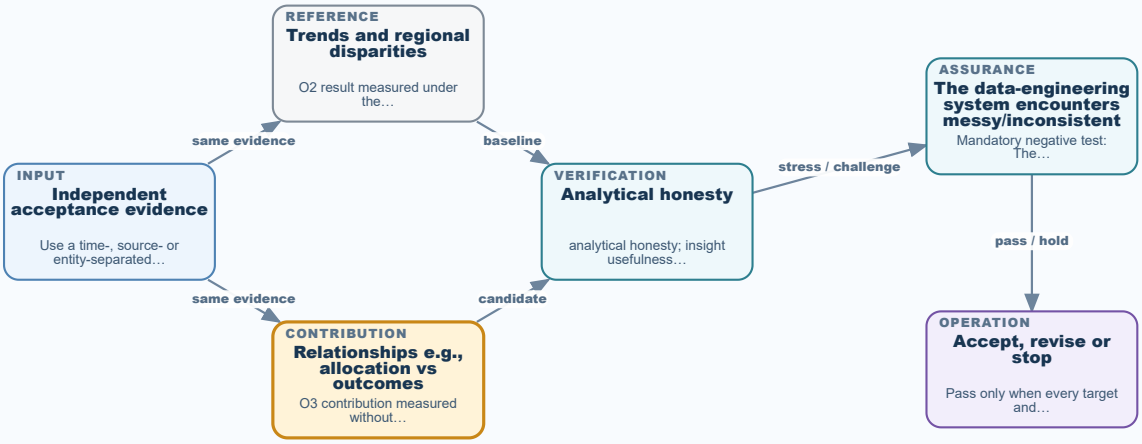
optimisation loop

A. System / physical architecture — what the team must build and connect



Every arrow is an interface or evidence transfer that must be implemented and tested.

B. Validation architecture — how the team must prove the result



Every arrow is an interface or evidence transfer that must be implemented and tested.

Two-semester execution roadmap

Semester VII: problem + reference · Semester VIII: contribution + system + acceptance

SEMESTER VII / CAPSTONE PROJECT I

O1

Validate the problem and freeze the engineering contract

Select and integrate public datasets, resolving quality issues. Verify the beneficiary, source claim, system boundary, evidence access, interfaces, hazards and acceptance rules before implementation proceeds.

CP1 Gate 1 — the problem, scope, evidence and acceptance plan are approved.

Mandatory outcome and evidence

An integrated, cleaned civic dataset. Required evidence: source-to-claim map, stakeholder or reproduction evidence, requirements, architecture decision record, risk register and source-to-result lineage, schema and time semantics, quality rules and idempotent replay evidence.

O2

Build the reproducible reference and vertical slice

Analyse trends and regional disparities. Compare at least two feasible reference approaches, select one through a documented trade-off and measure it under AC-1 and AC-4.

CP1 Gate 2 — a reproducible reference and an integrated vertical slice are demonstrated.

Mandatory outcome and evidence

Documented trends and disparities. Deliver executable fixtures, an independent oracle, unit tests, resource profile, failure log and baseline values for analytical honesty; insight usefulness; transparency of methodology; end-to-end lineage coverage (%).

SEMESTER VIII / CAPSTONE PROJECT II

O3

Engineer and prove the differentiating contribution

Analyse relationships (e.g., allocation vs outcomes) honestly. Compare it with O2 using the same evidence and resource envelope; include ablation, sensitivity, trade-off or controlled experiments appropriate to the project.

CP2 Gate 3 — the contribution is technically defensible and its trade-offs are measured.

Mandatory outcome and evidence

Defensible relationship analysis. Report baseline-versus-candidate results, uncertainty, condition-wise failures and a defensible positive or negative conclusion. Retain the selected configuration, experimental controls, raw evidence and a trace from each claim to its acceptance result.

04

Integrate the complete operable system

Build the public accountability dashboard with transparent methodology. Connect the evidence path, reference and contribution, user or operator interface, configuration, security or safety controls, telemetry, degraded mode and recovery.

CP2 Gate 4 — the complete system operates normally, degrades safely and recovers.

Mandatory outcome and evidence

A transparent civic-insight dashboard. Every component and interface must have unit, integration and system evidence in a representative environment. Retain configuration, interface contracts, telemetry, degraded-mode behaviour and recovery proof.

05

Complete independent acceptance and impact validation

Evaluate analytical honesty and dashboard usefulness. Execute AC-1 to AC-4 and all 5 negative tests; obtain independent domain, stakeholder or technical review.

CP2 Final Gate — every target and guardrail passes, or the project is explicitly held or revised.

Mandatory outcome and evidence

A validation report on rigor and usability. Publish baseline, target and achieved values, limitations, residual risks, reproducibility package, operating guide, final report, demonstration video and individual contribution evidence.

Functional requirements

- FR-1 — Ingest or acquire “Government open datasets data.gov.in / city portals”; validate version, provenance, schema or protocol, units and project-specific quality gates before use.
- FR-2 — Reproduce “Trends and regional disparities” through a one-command, one-protocol or one-test-run reference path with an independent expected-result oracle.
- FR-3 — Implement “Relationships e.g., allocation vs outcomes honestly” as a separately testable subsystem whose configuration and evidence are versioned.
- FR-4 — Implement and verify “Public accountability dashboard with transparent methodology” and its input, output, error, configuration and telemetry contracts.
- FR-5 — Implement and verify “Python / SQL analytics” and its input, output, error, configuration and telemetry contracts.
- FR-6 — Deliver “Civic Open-Data Dashboard for Public Accountability” as the complete demonstrable result; no isolated notebook, model, function or dashboard counts as completion.
- FR-7 — Detect the condition described by NT-1, invoke the documented safe or degraded response and retain recovery evidence.

Non-functional and domain-assurance requirements

- NFR-1 — Performance: meet every accepted KPI target under AC-1 and AC-4, including tail values and resource cost where applicable.
- NFR-2 — Reliability and recovery: pass every mandatory negative test; a critical failure cannot be hidden by an average.
- NFR-3 — Domain assurance: Every published result shall retain source-to-result lineage, schema and event-time semantics.
- NFR-4 — Evidence quality: Backfill, retry and replay shall be idempotent and reconciliation shall detect silent divergence.
- NFR-5 — Maintainability: use version control, automated tests, documented interfaces, configuration validation and a clean-environment build or protocol.
- NFR-6 — Operability: provide logs, measurements, health or quality indicators, operator guidance, degraded mode and a verified recovery procedure.

System boundary and permitted scope

INCLUDED

- Reproducible reference: Analyse trends and regional disparities. This is the O2 reference against which every Civic Open-Data Dashboard for Public Accountability contribution must be compared
- Differentiating contribution: Analyse relationships (e.g., allocation vs outcomes) honestly. This is the O3 technical contribution that must be implemented, isolated and evaluated separately
- Integrated result: Build the public accountability dashboard with transparent methodology. The final demonstration must show the complete Civic Open-Data Dashboard for Public Accountability workflow rather than an isolated component
- Independent acceptance: Evaluate analytical honesty and dashboard usefulness. The result must be independently checked under representative, boundary and failure conditions

EXPLICITLY EXCLUDED

- Production certification, universal generalisation and unattended high-stakes operation outside the validated envelope are out of scope.
- A notebook, isolated model, single-function prototype, dashboard-only demonstration or unvalidated library integration.
- Production certification, universal generalisation and unattended high-stakes operation outside the validated envelope.
- An isolated notebook, model, function, dashboard or library integration without the complete operating and validation workflow.

DEPENDENCY ASSUMPTION

Access to every required dataset, sample, device, tool, reviewer, licence and approval must be confirmed at the project-approval review; unavailable dependencies require an approved scope revision.

Team, prerequisites, tools and feasibility

RECOMMENDED TEAM

A team of 3–4 students must assign named responsibility for source data and lineage leadership, pipeline or storage engineering leadership, serving and observability leadership, reconciliation and data quality leadership in Civic Open-Data Dashboard for Public Accountability; each member must own reviewable implementation and evidence.

PREREQUISITES

Analytics, data integration, civic context.

PRINCIPAL TOOLS

- Python / SQL (analytics)
- data integration
- public dashboard
- versioned schema and fixture tools

FACILITY / COMPUTE

A reproducible local or cloud data platform with versioned fixtures, lineage, quality and replay tooling. Principal access must include Python / SQL (analytics), data integration, public dashboard, versioned schema and fixture tools.

FALLBACK PATH

Use privacy-safe synthetic or public fixtures with the same schema, time, quality and scale characteristics required by the problem. It may reduce scale but must retain O2, O3, O4 and O5.

Acceptance conditions

AC-1 **Representative operation**

Evaluate analytical honesty and dashboard usefulness.

AC-2 **Boundary and failure operation**

Exercise messy/inconsistent open data; easy to misread; avoiding misleading conclusions.

AC-3 **Independent acceptance evidence**

Use a time-, source- or entity-separated acceptance partition and reconcile every published result to source lineage.

AC-4 **Frozen resource envelope**

Use the same approved versions, workload, hardware or facility limits and resource budget for reference and final comparisons; principal tools are Python / SQL (analytics), data integration, public dashboard, versioned schema and fixture tools.

KPI baseline, target, method and pass contract

ID	MEASURE	BASELINE	DIRECTION	UNIT	TARGET / PASS RULE	CONDITIONS	MEASUREMENT METHOD	GUARDRAIL
KPI-1	**analytical honesty**	O2 reference measured with the same evidence, workload and resource envelope	higher is better	dimensionless score on a 0–100 rubric frozen before O2	≥ 80/100 and at least 5 points higher than the O2 reference	AC-1, AC-3	Before O2, freeze a 0–100 rubric with at least five observable criteria and named 0, 50, 80 and 100 anchors; use two independent raters and report agreement plus condition-wise scores.	No critical negative test, data-integrity failure or unapproved requirement regression is permitted.
KPI-2	**insight usefulness**	O2 reference measured with the same evidence, workload and resource envelope	higher is better	dimensionless score on a 0–100 rubric frozen before O2	≥ 80/100 and at least 5 points higher than the O2 reference	AC-1, AC-3	Before O2, freeze a 0–100 rubric with at least five observable criteria and named 0, 50, 80 and 100 anchors; use two independent raters and report agreement plus condition-wise scores.	No critical negative test, data-integrity failure or unapproved requirement regression is permitted.
KPI-3	**transparency of methodology**	O2 reference measured with the same evidence, workload and resource envelope	higher is better	dimensionless score on a 0–100 rubric frozen before O2	≥ 80/100 and at least 5 points higher than the O2 reference	AC-1, AC-3	Before O2, freeze a 0–100 rubric with at least five observable criteria and named 0, 50, 80 and 100 anchors; use two independent raters and report agreement plus condition-wise scores.	No critical negative test, data-integrity failure or unapproved requirement regression is permitted.
KPI-4	**end-to-end lineage coverage (%)**	O2 reference measured with the same evidence, workload and resource envelope	higher is better	%	≥ 95%; 100% for safety-, security- or compliance-critical items	AC-1, AC-3	Use versioned fixtures and an independent oracle; report condition-wise values, repeated trials and uncertainty.	No critical negative test, data-integrity failure or unapproved requirement regression is permitted.

ID	MEASURE	BASELINE	DIRECTION	UNIT	TARGET / PASS RULE	CONDITIONS	MEASUREMENT METHOD	GUARDRAIL
KPI-5	**reconciliation error count**	O2 reference measured with the same evidence, workload and resource envelope	lower is better	count	= 0 in every mandatory and adversarial condition	AC-2, AC-3	Execute every mandatory, boundary and adversarial fixture against an explicit expected-state oracle; retain complete logs and state evidence.	No critical negative test, data-integrity failure or unapproved requirement regression is permitted.
KPI-6	**pipeline completion latency p95 (s)**	O2 reference measured with the same evidence, workload and resource envelope	lower is better	s	p95 ≤ 0.80 × O2 reference at equal or better functional quality	AC-1, AC-4	Warm the system, run at least 30 measured trials per condition with a monotonic clock, and report p50, p95, p99 and failures.	The primary correctness, safety or quality metric may not regress by more than the approved non-inferiority margin.

Mandatory negative-test and recovery campaign

NT-1 **The data-engineering system encounters messy/inconsistent open data during the Civic Open-Data Dashboard for Public Accountability operating or validation workflow.**

Trigger: Create a controlled fixture, sample, workload or operating condition that reproduces this failure: “The data-engineering system encounters messy/inconsistent open data during the Civic Open-Data Dashboard for Public Accountability operating or validation workflow”.

Expected safe behaviour: The invalid record or state transition must be detected before silent commit; replay must be idempotent and lineage must remain complete.

Recovery: Quarantine invalid data, repair from the versioned source or checkpoint, replay idempotently and reconcile every downstream result.

Evidence: Retain fixture or sample ID, precondition, version and configuration, expected versus observed result, logs or measurements, recovery time and residual-risk decision.

NT-2 **The data-engineering system encounters easy to misread during the Civic Open-Data Dashboard for Public Accountability operating or validation workflow.**

Trigger: Create a controlled fixture, sample, workload or operating condition that reproduces this failure: “The data-engineering system encounters easy to misread during the Civic Open-Data Dashboard for Public Accountability operating or validation workflow”.

Expected safe behaviour: The invalid record or state transition must be detected before silent commit; replay must be idempotent and lineage must remain complete.

Recovery: Quarantine invalid data, repair from the versioned source or checkpoint, replay idempotently and reconcile every downstream result.

Evidence: Retain fixture or sample ID, precondition, version and configuration, expected versus observed result, logs or measurements, recovery time and residual-risk decision.

NT-3 **The data-engineering system encounters avoiding misleading conclusions during the Civic Open-Data Dashboard for Public Accountability operating or validation workflow.**

Trigger: Create a controlled fixture, sample, workload or operating condition that reproduces this failure: “The data-engineering system encounters avoiding misleading conclusions during the Civic Open-Data Dashboard for Public Accountability operating or validation workflow”.

Expected safe behaviour: The invalid record or state transition must be detected before silent commit; replay must be idempotent and lineage must remain complete.

Recovery: Quarantine invalid data, repair from the versioned source or checkpoint, replay idempotently and reconcile every downstream result.

Evidence: Retain fixture or sample ID, precondition, version and configuration, expected versus observed result, logs or measurements, recovery time and residual-risk decision.

NT-4 **Data quality is measured globally while a critical partition remains defective in Civic Open-Data Dashboard for Public Accountability.**

Trigger: Create a controlled fixture, sample, workload or operating condition that reproduces this failure: “Data quality is measured globally while a critical partition remains defective in Civic Open-Data Dashboard for Public Accountability”.

Expected safe behaviour: The invalid record or state transition must be detected before silent commit; replay must be idempotent and lineage must remain complete.

Recovery: Quarantine invalid data, repair from the versioned source or checkpoint, replay idempotently and reconcile every downstream result.

Evidence: Retain fixture or sample ID, precondition, version and configuration, expected versus observed result, logs or measurements, recovery time and residual-risk decision.

NT-5 **The data-engineering system encounters access controls or retention rules expose governed or sensitive evidence in Civic Open-Data Dashboard for Public Accountability during the Civic Open-Data Dashboard for Public Accountability operating or validation workflow.**

Trigger: Create a controlled fixture, sample, workload or operating condition that reproduces this failure: “The data-engineering system encounters access controls or retention rules expose governed or sensitive evidence in Civic Open-Data Dashboard for Public Accountability during the Civic Open-Data Dashboard for Public Accountability operating or validation workflow”.

Expected safe behaviour: The invalid record or state transition must be detected before silent commit; replay must be idempotent and lineage must remain complete.

Recovery: Quarantine invalid data, repair from the versioned source or checkpoint, replay idempotently and reconcile every downstream result.

Evidence: Retain fixture or sample ID, precondition, version and configuration, expected versus observed result, logs or measurements, recovery time and residual-risk decision.

Evidence and reproducibility contract

PRIMARY EVIDENCE

government open datasets (data.gov.in / city portals)

PROTOCOL AND QUALITY

Version the cited evidence and project inputs, reproduce “Analyse trends and regional disparities”, implement “Analyse relationships (e.g., allocation vs outcomes) honestly”, and compare both under AC-1 to AC-4 without changing the approved evidence or resource envelope. Check schema, null/duplicate rules, event time, source-to-output lineage, replay consistency and reconciliation totals.

PROVENANCE AND INDEPENDENT PARTITION

The Civic Open-Data Dashboard for Public Accountability evidence manifest must record source URL, access date, licence or permission, owner/custodian, record/sample/fixture identifiers, exclusions, transformations, configuration and the objective or KPI using each item. Reserve the AC-3 evidence before O3 tuning and separate it by the independent unit appropriate to data-platform engineering; no transformation may leak acceptance information into development.

REPRODUCIBILITY

A clean environment must rebuild the O2 reference, O3 contribution, integrated system and acceptance evidence for Civic Open-Data Dashboard for Public Accountability using the version-pinned toolchain (Python / SQL (analytics), data integration, public dashboard, versioned schema and fixture tools), seeds/configuration and an automated or fully scripted runbook.

Mandatory final deliverables

- D1 — Approved Civic Open-Data Dashboard for Public Accountability problem charter, requirements, system boundary, architecture and source-to-claim map.
- D2 — Reproducible Trends and regional disparities with fixtures or protocol, oracle, tests, baseline results and resource profile.
- D3 — Independently testable Relationships e.g., allocation vs outcomes honestly with design rationale, comparative evidence, trade-offs and limitations.
- D4 — Complete Civic Open-Data Dashboard for Public Accountability prototype or experimental system with configuration, interfaces, telemetry, degraded operation and recovery.
- D5 — Automated or documented acceptance harness covering AC-1 to AC-4, all KPIs and every negative test.
- D6 — Versioned repository or protocol package, evidence manifest, installation/operating guide, API/interface documentation and reproducibility instructions.
- D7 — Final report, research paper or technical article, poster, portfolio case study, demonstration video, video logs and individual contribution/viva evidence.